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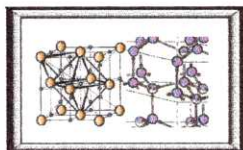
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COLD FUSION TIMES

de novo NUCLEOSYNTHESIS JOINS EXCESS HEAT

Mitchell Swartz

College Station, Texas -- This issue of the **COLD FUSION TIMES** features several articles reporting evidence consistent with nucleosynthesis. Most compelling is the appearance of new isotopic distributions not previously reported. These accounts are included along with detailed reviews of The Second International Low Energy Nuclear Reactions Conference (ILENR2) which was held on September 13-14, 1996 at College Station. Some of the transmutations, which was the predominant topic of the meeting, appear to occur in certain fully loaded Group VIII transition metals. Others, purportedly occur in explosive inorganic mixtures. Several of the presentations have again confirmed developing evidence that the simplicities of Michael Faraday (1791-1867) and Walther Nernst (1864-1941), although true in many ways, do not fully define the reactions or the new physics/chemistry/material-science involved when certain metals are loaded to high densities. Today, science rests on the shoulders of these early researchers, but they may have underestimated

Cont. page 2

Heat and Nuclear Products by Cavitation Induced Fusion

Russ George

Mountain View, California - E-Quest Sciences has developed a new energy technology using ultrasound induced cavitation in heavy water upon specified metal target foils which results in energy (heat) production from a here-to-fore unknown class of nuclear reactions. E-Quest's micro-fusion employs the use of ultrasound (20 khz) induced cavitation to provide an environment where hydrogen isotopes might combine with a target lattice and produce a similar nuclear reaction.

Experiments using this technology have produced robust excess heat (resultant power up to 60+% greater than the input power) and nuclear products (principally but not exclusively ^4He). Monitoring of similar previous experiments with various gamma and neutron sensitive equipment revealed no energetic emissions above ambient background levels.

Cont. page 6

2ND INTERNATIONAL TRANSMUTATION CONFERENCE -- ADDITIONAL EVIDENCE CONSISTENT WITH NUCLEOSYNTHESIS



Leaders in transmutation and nucleosynthesis research from left (Drs. John Bockris, Hal Fox, Tadahiko Mizuno, George Miley). More photographs and information on pages 2, 6, 7, and 9.

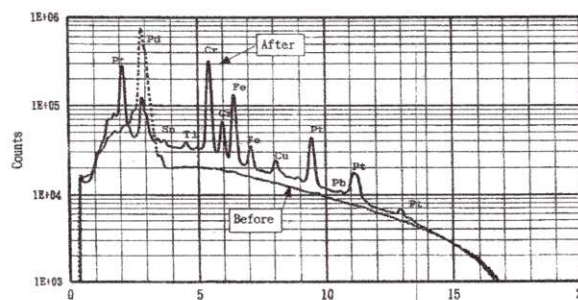
Anomalous Isotopic Distribution in Palladium Cathodes after Electrolysis

Tadahiko Mizuno, Tadayoshi Ohmori, Michio Enyo

Sapporo, JAPAN -- Nuclear reactions in a solid electrode at ordinary temperatures have been reported by many experimenters since 1989. However, this phenomenon is still not well accepted among researchers because of poor reproducibility and control. What is urgently needed now is to obtain precise and

terms of the nuclear mechanisms. In this work, evidence which indicates the occurrence of some nuclear reactions is presented, in the form of transmuted elements within the cathode and on the cathode surface. The anomalous isotopic distribution of these elements shows they do not come from contamination.

Cont. page 5



EDX Spectra From A Palladium Rod Before And After Electrolysis
Energy dispersive X-ray spectroscopy (EDX)
The evolution of Pt, Cr, Fe, Cu and other elements are seen (after Mizuno)

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TOWARD THE FUTURE

Due to information-dense coverage of the ILENR2 Conference, the COLD FUSION TIMES's "TOWARD THE FUTURE" will reappear next issue. These are important excerpts of Jed Rothwell's complete article on ILENR2, which is available in *Infinite Energy* #9.

Summary on ILENR2

Jed Rothwell



College Station, Texas - This conference was held on September 13-14, 1996 at the Holiday Inn, College Station, Texas, adjacent to Texas A&M University, because it was banned by the university. It was chaired by Distinguished Professor J. O'M Bockris of Texas A&M and G. Miley, University of Illinois. The conference was organized by Dr. G. H. Lin. The focus was transmutations of elements in the cathode (so-called "host metal transmutations"), rather than excess heat, neutrons and other well-known attributes of the cold fusion effect. Most papers described transmutations in conventional electrolysis cold fusion experiments, but other approaches were discussed, including ultrasound loading; arc discharge synthesis; and some variations of ancient alchemical techniques. I consider this one of the most important meetings in the history of cold fusion. These results constitute a major turning point in the search for nuclear ash and a mechanism. I consider it now proven beyond doubt that the Pons-Fleischmann effect often (perhaps always) causes both fission and fusion of heavy elements in the cathode. It is nothing like a light element plasma fusion reaction.

Miley [1] This was the most outstanding paper, and one of the best presentations I have

ever seen. Miley began by describing his previous multilayer thin film flat plate reported on at ICCF3. As he said it was a great success insofar as the thin film heated up anomalously and peeled off. He described this as a good example of what Edison called "learning 700 things that don't work." Miley personifies the good humored, dogged persistence required to solve big problems like cold fusion. As he said, he started out a plasma physicist in 1989 and he has learned every instrument and technique in cold fusion.

He is working CETI-style thin film beads packed in 3 to 5 layers. The beads are fabricated by the Material Science Department at the university. They have devised a new type of sputtering machine that shakes the beads, levitating and rotating them to produce less than 10% deviation in coating thickness. Miley says the sputtering technique simplifies the problem because it does away with the need for a copper flashing, reducing the number of elements at the start. In conversations he has also told me that he thinks the coating is tougher and the beads will generate heat more readily than Patterson's electrodeposited beads. Miley's has made a great effort to prevent contamination in the closed, flow calorimeter. He describes contamination as his greatest worry. He has performed 20 runs to date. He has performed the most extensive analysis on run #8, done in March 1996. The analysis is "very tedious" taking weeks of hard work. He presented voluminous comparative charts comparing data from different runs. He uses four techniques to measure elements and isotopes: NAA, SIMS, EDS and AES.

Cont. page 9

de novo NUCLEOSYNTHESIS

Continued from page 1

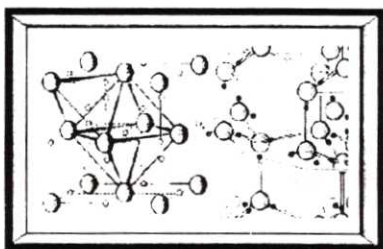
the full nature of conduction processes and the potential absence of an equilibrium condition. A century and a half later, we now know that the structure of water, hydrided electrodes and even their interfaces are a much more complicated matter. Nucleosynthesis joins excess heat (enthalpy) as two more proven, but difficult to accept, intraelectrode phenomena occurring only when active samples are loaded to very high loading levels. Several excellent reports include those by Drs. Mizuno and Bockris.

Dr. T. Mizuno and his coworkers at the Dept. of Nuclear Eng. at Hokkaido Univ., Sapporo Japan, studied cathodic electrolysis

followed by a careful spatial examination of the isotopic element analysis, indicates that there appears to be very low level nucleosynthesis in such samples. They have utilized several methods of analysis which have indicated the presence of reaction products with mass numbers ranging from 39 to 81, 104 to 136, and 177 to 208 following electrolysis upon palladium cathodes in a heavy water solution at high pressure, high temperature, and high current density for one month duration. Perhaps the most remarkable feature is that the isotopic distributions were radically different from the natural ones.

Drs. Lin and Bockris, from Texas A&M University's Department of Chemistry, have conducted novel investigations involving

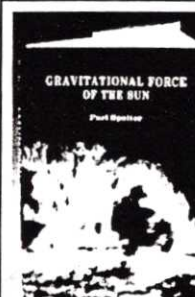
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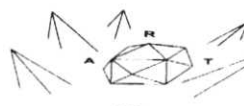


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Page 2: Jed Rothwell - JET Technology

Page 3: Barry Merriman (this web page tk)

Page 6: Russ George - JET Technology

Page 9: ILENR2 Meeting - JET Technology

Back Cover: Palladium microcrystal

Cold Fusion Times LOGO - Palladium must be fully filled with deuterons to generate the reactions. In the logo, the face-centered cubic fully loaded metallic structure is shown on the left, totally filled with deuterons obtained from the heavy water on the right hand side.

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PEOPLE IN THE NEWS

A Critical Report on ILENR2

Barry Merriman



I too was at the conference. I am too busy to give a detailed account, but Jed has done a good job of recounting most of the interesting presentations.

But, let me make a few comments. First, the organizers allowed a several people to speak who had no business presenting at any sort of scientific meeting. Anyone present can readily identify those folks, ho particularly dominated the entire afternoon session. That they were allowed to speak is an act of charity that denigrates the serious work of Miley, Mizuno, etc. --- not to mention that these researchers could have used much more than 25 minutes to present their quite interesting results. In future meetings, they really need to exercise quality control.

Second, I think that I and Dr. Natowitz, head of the university Synchrotron Department, were the only non-believers present. At times, we both voiced the obvious objections about contamination, isotope separation via diffusion through the metal lattice, etc. I don't know if Natowitz used these objections to dismiss he results or not (I was out of the room for 20 minutes during the final 2 hour discussion period having my own discussion, so maybe I missed something.)

Personally, I feel these objections do not _obviously_ explain the results. They are simply possibilities that must be addressed. On the face of it, they would appear inadequate, since Miley, Mizuno and Ohmori were attempting to take great care to prevent contamination, and since the distribution of elements is too deep inside the metal lattice to get their by classical diffusion. BUT, on the other hand, none of the researchers has yet found a

good blank (because they see effects with all metals and both heavy and light water (?!?), and they have done only limited controls as well at this point. Also, for example, Miley's setup was run without a peristaltic pump (!), so there was some metallic contact with the electrolyte. (It is inconceivable to me why they would do this, but it seems o have been a historical artifact and a budgetary limitation according to Miley). Rather than dismiss the results as Jed claims Natowitz is prone to do, all I would say is that they are extremely interesting, and at this point what is needed is to establish rigorous, detailed control experiments --- not blanks, since they seem to not exist --- so that one can quantify the background effects. I spent a great deal of time discussing possible controls with Miley, and I feel there are controls that would be adequate to prove the reality of the effect. The possibilities to watch out for are unanticipated contamination, and perhaps some form of anomalous diffusion of isotopes through metals. (Not as bizarre as it may seem: one Russian fellow in the audience mentioned that in plasma ion implantation, one does see huge anomalous transport of ions through the lattice, since the resulting lattice defects act as carriers, and defects are mobile. Also, in the hot fusion world, if it were not for the anomalous diffusion of deuterium through the plasma material, we would have had cheap, working hot fusion reactors 30 years ago. The possibility of some form of greatly enhanced diffusion cannot be lightly dismissed.)

Bottom line is: very interesting, but I want tighter controls. I have a lot of respect for Miley in particular, since I have known him for years from his hot fusion research, where he has consistently done innovative, quality work. I am confident the necessary controls will be done; and we will know whether it is transmutation or contamination + anomalous diffusion.

Barry Merriman is a Research Scientist with UCSD Fusion Energy Research Program, and Asst. Prof., UCLA Dept. of Math and offers constructive criticism about the conference.

Production of Elements and Plasmoids Edward Lewis

Plasmoid research goes back decades. W. Bostick produced things he called "plasmoids" by discharging through electrodes, and in 1957 he speculated (1) about the identity of galaxies and particles as "plasmoids." According to A. Peratt (2), Bostick coined the term "plasmoid." From Bostick's early work, and Alfven's early work with plasmas, a whole school of plasma astrophysical research also has developed. E. Lerner published (3) about the development of astrophysical plasmoid theories. The plasmoid research also led to some ball lightning research and modeling. Recently, Lewis resolved the phenomena of ball lightning and the tiny objects emitted during the cold fusion phenomena as plasmoid phenomena, and developed a general theory about phenomena as plasmoid phenomena. The various kinds of plasmoids emitted by the electrolysis and discharge apparatus used by Matsumoto emit tiny plasmoids which he has photographed and leave several kinds of traces on emulsions. They travel through glass, water, and air to get to the emulsions, which is a behavior like that of ball lightning.

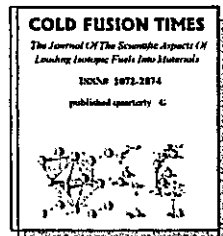
There are two different kinds of holes people find in electrodes: round ones due to plasmoid boring and intergrain

or grain-shaped voids that may be lined with new elements and crystals that are due to the clumping or conversion of plasmoid atoms. Liaw and Silver (4) have photographed such seemingly anomalous micrometer size round pits or produced SEM pictures of them. Silver and Matsumoto (5) have shown photographs or SEM pictures of micrometer size ditches or trails in their electrodes that are similar to the trails on the nuclear emulsions that Matsumoto has shown. K. Shoulders has reported both pits and tunnels, and ditches or grooves (6) in materials produced by the plasmoids he uses. Interestingly, the new element filled voids in the electrode that Matsumoto used seem to be "near the surface" or opening on the surface, almost entirely within about 20 micrometers of the surface, most within 10 micrometers. But the voids are quite large, sometimes greater than 8 micrometers in width. In the same way, the voids in Numata et al.'s (7) electrode seem to be on and near the surface, in a layer of grains at the surface distinct from the two huge crystals that make up the bulk.

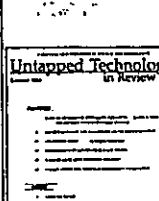
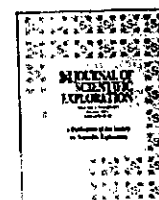
1. W. Bostick, "Plasmoids," Scientific American, 197, 87 (October 1957).
2. A. Peratt, email note, January 27, 1995.
3. E. Lerner, The Big Bang Never Happened. Vintage Books, New York, 1991, 1992.
4. D. Silver, J. Dash, and P. Keefe, "Surface Topography of a Palladium Cathode After Electrolysis in Heavy Water," Fusion Technology, 24, 423 (December 1993).
5. T. Matsumoto, "Experiments of One-Point Cold Fusion," Fusion Technology, 24, 332 (Nov. 1993).
6. K. Shoulders, "Energy Conversion Using High Charge Density," Patent Number 5,123,039.
7. H. Numata, R. Takagi, I. Ohno, K. Kawamura, and S. Haruyama, "Neutron Emission and Surface Observation During a Long-Term Evolution of Deuterium on Pd in 0.1 M LiOD," Proceedings of the 11 Annual Conference on CF, T. Bressani et al. editors, Como, Italy, June 29 - July 4, 1991.

OF GREAT INTEREST
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3



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**Journal Articles Which
You May Have Missed**
(Courtesy of Dieter Britz)

Arata Y, Zhang Y-C; *Koon Gakkaishi* 21(6) (1995) 303 (in Japanese, Engl. abstr.). "Achievement of solid-state plasma fusion ("cold fusion").** Experimental, gas phase Pd, D₂, mass spec, helium, excess heat. This pair of authors has been producing a steady stream of papers, using their Pd powder in a Pd bottle, and D₂ gas, and producing excess heat. They now add a high-resolution mass spectrometer. They repeatedly scan for masses between 3.95 to 4.05, and find a distinct ⁴He peak at 4.00260 appearing next to that for D₂, 4.02820, under those conditions where they see excess heat and claim cold fusion. No ³He or tritium was detected. There are controls, and the He appears only when heating the Pd sufficiently, indicating that it is produced within the metal.

Celani F, Spallone A, Tripodi P, Petrocchi A, Di Gioacchino D, Boutet A, Marini P, Di Stefano, Diociaiuti M, Collis W; *Fusion Technol.* 29 (1996) 398. "Reproducible D/Pd ratio >1 and excess heat correlation by 1-microsec-pulse, high-current electrolysis".

A flow calorimeter was used, and the cathode was a number of 25*25*1 mm³ Pd sheets, with surrounding Pt anode, in 0.3 M LiOD. Loading was driven by high-level (100 A peak), short-time (< 300 ns) current pulses corresponding to a mean current of 64 mA. Loading was measured by the deficit in the gases released and checked by weighing the cathodes afterwards, heating to drive out the deuterium and reweighing; the two measures agreed. Checks with nonabsorbing Au cathodes showed an error of only 0.01 in loading. Loading time was about 3 days and loadings well above 1 -up to 1.2- were achieved. Excess heat, calculated after correction for the heat of water electrolysis, was up to nearly 100%, but not reproducible, possibly due to surface effects. The absorption rate seemed to be an indicator of success.

Wang D, Chen S, Li Y, Liu R, Wang M, Fu Y, Zhang X, Zhang W; *Yuanzi Yu Fenzi Wuli Xuebao* 10(3) (1993) 2789 (in Chinese). Cited in Chem. Abstr. 120:228892 (1994).

"Neutrons, gamma-rays and x-rays in a gas discharge". "An anomalous phenomenon in a metal loaded with D was studied by using the gas-discharge method, and ca. 10⁴ n/s were detected. The prodn. of n is controllable and repeatability is 100%. Neutrons at energies of 1.0-3.0 MeV were measured by the NE213 detector. Anomalous x-rays of av. energy (27.6^{+2.1}) keV were measured. Anomalous gamma-rays of energy ca. 470 keV were measured."

"Cold nuclear fusion apparatus", Toichi Chikuma, JP 07,287,085 (95,287,085), 31 Oct. 1995, 18 Apr. 1994. Apparatus equipped with a means around an absorbent (such as Pd or ceramic) which absorbed a material (such as D) which starts the nuclear fusion to prevent the absorbed material from escaping from the absorbent. The means is a magnetic coil. The means may comprise an absorbent (which is a cathode), an electrical conductive layer (which is used as an anode) is formed around the absorbent via an electrical insulator, and voltage is applied. The efficiency of the nuclear fusion is improved.

"Cold nuclear fusion apparatus", Megumi Myake, JP 07,244,176 (95,244,176), 19 Sept. 1995, 4 Mar. 1994. Apparatus is composed of putting D-absorbing Pd in a D bath and highly heat conductive Cu is connected to it.

"System for Electrolysis", James A. Patterson, US 5,494,559, 27 Feb. 1996, 8 Jun. 1995; 52 claims.

A system for electrolysis and heating of a liquid electrolyte comprising: an electrolytic cell including a non-conductive housing and an inlet and an outlet; a first conductive foraminous grid positioned within said housing adjacent to said inlet; a second conductive foraminous grid positioned within said housing spaced from said first conductive grid and adjacent to said outlet; a plurality of conductive microspheres of substantially uniform size and density in electrical communication with said first conductive grid and electrically isolated from said second grid; said plurality of microspheres each including: a conductive metal flash coating of uniform thickness formed by chemical combination with a cation exchange surface of a spherical cross-linked polymer microbead from a metal cation which will chemically reduce with hydrazine; a nickel plating of uniform thickness formed atop said flash coating; a metallic hydride forming plating of uniform thickness formed atop said nickel plating, said metallic hydride plating being readily combinable with hydrogen or an isotope of hydrogen; a metallic support plating of uniform forming plating, said support plating having a relatively high hydrogen diffusion rate and a relatively low hydride formation ratio; means for pumping said liquid electrolyte into said electrolytic cell through said inlet, said electrolyte having a conductive salt in solution with water, said electrolyte exiting from said electrolytic cell through said outlet; said pumping means maintaining said electrolytic cell substantially filled with said electrolyte; an electric power source having terminals operably connected to said first and second grids.

"Porous metal cathode for cold fusion chain reaction, its manufacture, and electrolyte for reaction", Reiko Notoya, JP 07,174,878 (95,174,878), 14 July 1995, 29 Oct. 1993, The cathode consists of a porous metal (voids 0.5-80 volume %) selected from a transition metal,

Al, Sn, and stainless steel. The cathode is manufactured by shaping a metal powder with grain size 10 nm -100µm at room temperature or high temperature and high pressure, or shaping and heating. The electrolyte comprises a hydroxide, carbonate, sulfate, phosphate, nitrate, halide, perchlorate, and/or B compound of an alkali metal, alkaline earth metal, Group IIIB elements or transition metals.

"Generation of thermal energy by cold nuclear fusion", Arata Y; Jpn. Kokai Tokkyo Koho, JP 08 75,882 [96 75,882], Sep-94. Cited in Chem. Abstr. 124:327077 (1996). Ultrafine Pd powder is put in a stainless steel vacuum vessel, then the vessel is filled with D₂ gas with pressure, the vessel is cooled with a cryostat to liquefy the D₂, the liq. D₂ is absorbed by the Pd until the Pd is satd. with the liq. D₂, then the temp. of the vessel is raised to a proper temp. to produce thermal energy based on cold nuclear fusion. Cold fusion can be carried out in a short time".

"Pulsed electrochemical converter of nuclear fusion energy and its operation", Arshinov AN, Burtsev AB, Gireva LK, Lisitsyn AL, Stankov VKh, Chizhik SP (Tovarishchestvo s Organichennoi Otvetstvennostyu "Khimelektro", Russia); Russ. RU 2046464 C1, Appl. 24-Nov-92. Chem. Abstracts 124:187771 (1996).

"Method for producing excess energy and neutron radiation in the mixture of the light atoms solved in current-carrying material immersed in electrolyte and/or of the isotopes of said atoms, respectively in the interfacial double layer of the electrolyte", Daruhazi L, Pinter OH; Hung. Teljes HU 71,652, Sep-93, cf. Chem. Abstr. 124:326752 (1996). * "A claim is made for cold fusion of light atoms in the interfacial double layer of an electrolyte. The electrodes are placed at min. 10 mm and immersed in the electrolyte". (from CA).



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X-Ray, Heat, X-ray and ⁴He in the Electrochemical Conversion of Deuterium in Palladium (Dr. Chizhik S. (Italy))
Heat and Palladium Cathodes of D₂O Electrolysis Cells Involving Excess Power (Dr. Jorgensen, S. (Denmark))
Achievement of Solid-State Plasma Fusion ("Cold Fusion") (Dr. Arata Y. (Japan))
Heat and Helium Measurements Using Palladium and Palladium Alloys in Heavy Water (Dr. Miles, M. H. (U.S.A.))
Further Measurements on ⁴He Production from Pd/D₂ Systems in Gas Phase (Dr. Botta, E. (Italy))
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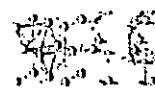
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MATERIALS CORNER UNUSUAL ISOTOPES APPEAR SUGGESTING NUCLEOSYNTHESIS

Anomalous Isotopic Distribution

T. Mizuno Continued from page 1

For example, natural copper is 70% Cu⁶³, and 30% Cu⁶⁵. But the copper found in the cathode was 100% Cu⁶³, with no detectable levels of Cu⁶⁵. Natural isotopic distribution varies by less than 0.001% for copper.

Palladium rods used were of high purity (99.97% min.) supplied by Tanaka Noble Metals, Ltd. Impurities in the sample were as follows; B: 110 ppm, Si: 10. Ca: 9, Cr: 10, Cu: 6, Ti: 5, Ag: 44, Mg: 1, Pt: 20 and Au: 23. Palladium rods used were of high purity (99.97% min.) supplied by Tanaka Noble Metals, Ltd. Impurities in the sample were as follows; B: 110 ppm, Si: 10. Ca: 9, Cr: 10, Cu: 6, Ti: 5, Ag: 44, Mg: 1, Pt: 20 and Au: 23. Nothing more was detected by atomic absorption photospectroscopy. Heavy water was supplied by Showa Denko, Ltd. It is 99.75% pure and includes 0.077 [micro]Ci/dm³ of tritium. The heavy water was purified once in a quartz glass distiller. Reagent grade lithium hydroxide was obtained from Merck, Ltd. Impurities in the reagent were specified as follows; Li₂CO₃: 2% max, Cl: 0.05%, Pb: 20 ppm, Ca: 200, Fe: 20, K: 200 and Na: 200. The anode and recombining catalyst were, respectively, a high purity (99.99%) Pt plate and a Pt mesh. The Pt metal is specified to contain impurities as follows; Rh: 18 ppm, Si, Cr and Pd: 2 ppm, Au, Ag, B, Ca, Cu and Fe; less than 1 ppm. Other impurities were under the limits of detection. Electrolysis was performed in a closed cell made of stainless steel. The cell inner wall was coated with 1 mm thick Teflon. The details have been described elsewhere (1). Before the electrolyte was added to the Pd cell, it was pre-electrolyzed with Pt electrodes at 1 A and 150 deg C for 6 x 10⁵ s (7 days). Electrolysis experiments were performed at a current density of 0.2 A cm² or total current of 6.6 A at 10⁵ n for 2.76 x 10⁶ s (32 days). The sample electrodes were analyzed for element detection by energy dispersive X-ray spectroscopy (EDX), Auger electron spectroscopy (AES), secondary ion mass spectroscopy (SIMS) and electron probe microanalyzer (EPMA).

Cont. page 7

de novo NUCLEOSYNTHESIS

Cont. page 5

nuclear transformations. Their research indicates anomalous radioactivity and the *de novo* appearance of unexpected elements. In some of their experiments this is a result of igniting a mixture containing inorganic compounds.

COLD FUSION TIMES features excerpts of reports from the ILENR2 meeting. Barry Merriman and Jed Rothwell (and Bob Horst) have been very helpful by summarizing what many felt was a good, solid meeting on a subject which appears to be developing compelling data.

In conclusion, what is most interesting about some of these reports is the monotonic increasing depth-related alterations in the generated isotopic distributions within Dr. Mizuno's electrodes. The volume integral of available all isotopes (e.g. contamination, corrosion materials) remains to be further clarified on a semiquantitative basis. However, when it is done, this new nucleosynthetic technology will have its own convincing demonstration that excess enthalpy and helium-4 production are now joined by generation of *de novo* elements in certain fully loaded active metals.

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Anomalous Radioactivity and Inorganic Mixtures

Lin and O'M. Bockris

Continued from page 10

Some methods have given more precious metals than others. It has been suggested that recovery of more gold from the ores by use of different fire assaying technique is the result of transmutation at chemical temperatures of other elements to gold instead of by means of a greater efficiency in extraction. In this paper, we report experimental results on anomalous radioactivity and appearance of small quantities of unexpected elements achieved by rapidly igniting a mixture of inorganic compounds. The details of the experiments and the results analyzed in our laboratory and in other private laboratories are presented.

Anomalous Radioactivity - A decay of β -radiation after igniting a chemical mixture has been observed [11]. The experiment was performed on April 14, 1992. The chemical composition of the mixture used are listed as follows: C (100 g), KNO₃ (300 g), S (30 g), SiO₂ (30 g), FeS (40 g), Hg₂Cl₂ (10 g), PbO (10 g). The chemicals were weighed separately, and then mixed together. The mixture was ignited in a hood. The burn lasted about 60 sec. The product was ground in a mortar with a pestle after cooling down. Samples from both the original mixture and the product were examined. Radioactive emission from the samples was detected by means of a detector (Model 44-1, Ludlum Measurements, Inc.), and recorded by a multichannel pulse height analyzer (Model 800, The Nucleus Inc.) and a data system (Model ZVM-123-A, Zenith). The detector was covered by a black box to shield it from the room light. The counts of the product were higher than those of the raw material, about 1.4 times greater. The counts of the raw material were stable. On the other hand, decay of the radiation from the product was observed. The β count of the product (background counts subtracted), as a function of time had exponential decay, and a half life time of 17.7 hours was calculated. Several other experiments were performed. Not every experiment gave the same results, i.e., conditions for the reproduction of the experiment are not yet well defined.

Unexpected Elements - Six experiments, named thermal 1 to thermal 6, were performed in Texas A&M University from April 30, 1992 to June 15, 1992. Four experiments showed an enhancement of gold by 1-2 orders of magnitude above the maximum concentration of gold present in the starting material. These results were confirmed by multiple independent analyses using several different methods and executed by various organizations. However, it was not found possible to reproduce these results in later attempts. Two decay experiments showed a half life time of around 20 hours. However, these two runs did not give positive gold production. In 1995, Noninski [12] suggested that the apparent decay of the burned product may have been caused by a layer of water adsorbed increasingly from the atmosphere. Thus, further investigation would be required for obtaining a definite conclusion as to the reality of the radioactivity. The lack of repeatability of the results firstly obtained, does not permit a conclusion that the examined "gunpowder" method caused transmutation at low energies. There is a tradition among mining engineers that, by use of such explosive techniques, an order of magnitude increase in the noble metals extracted from ores otherwise classed as "low grade" can be achieved. However, three successful experiments used pure chemicals, not

ores, as starting materials.

1. E. Rutherford, Nature, 103 (1919) 415.
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11. G. H. Lin, R. Bhardwaj, and J. O'M. Bockris, J. Sci. Exploration, 9, No. 2, 5 (1995).
12. V. C. Noninski, J. Ciottone, and P. White, "Experiments on Claimed Emission", J. Sci. Exploration 9, No. 3 (1995) 1.

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COLD FUSION TECHNOLOGIES

ACTIVE MEDIA? POSITIVE FEEDBACK?

M. Swartz, "Catastrophic Active Medium Hypothesis of Cold Fusion", Vol. 4, Proceedings: "Fourth International Conference on Cold Fusion", sponsored by EPRI and the Office of Naval Research, December (1994).

The catastrophic active medium (CAM) theory hypothesizes a two-stage system for cold fusion that ends with the production of the cracks which absorb and dissipate, rather than create, the energy required to activate the desired reactions. Following adequate deuteron loading, a local inhomogeneity of heat distribution in the metal is hypothesized to cause an in-situ fractional desaturation of the fully-loaded palladium. This desaturation is facilitated by optical phonons, which couple with deuteron transport. The CAM theory

of cold fusion hypothesizes that these desired reactions occur at select vacancies and defects.

Catastrophic deuteron, plasmon, phonon, and polaron fluxes are coupled with further exothermic deuteron desaturation to create an active medium and at least one positive feedback loop. There is possible internal conversion of any potential fusion reactions to the lattice by coupling through the phonons, already present from the desaturation, and polarons which increase the effective mass of the deuterons. The dynamic instability continues either until the active media is drained or, by a second catastrophic process, the fusion-defect-site is no longer confined.

S. Pons, M. Fleischmann (Cent. Sci., IMRA Europe SA, Valbonne, France), "Calibration of the Pd-D₂O System: Effects of Procedure and Positive Feedback," *J. Chim. Phys. Phys.-Chim. Biol.*, vol 93, no 4, (1996), pg 711-730.

The authors outline some of the considerations which have prompted their research on anomalously fast nuclear reactions of D⁺ compressed electrochem. into Pd (and Pd alloy) host lattices. The most surprising result has been that the generation of high levels of excess enthalpy is not accompanied by the expected levels of tritium and

neutron generation. Some of the major steps in the development of this particular aspect are outlined; it has been found that excess heat production is dependent on the protocol of the experiments mainly because of "positive feedback." A rationale for such "positive feedback" is presented which also explains oscillations in the system properties. The authors illustrate the progressive development of the investigation leading to the achievement of specific rates of excess enthalpy production of 4 kW cc at temperatures up to 100 degrees (i.e. of low-grade heat).

REVIEWS OF: Milton, Richard; "Forbidden Science. Suppressed research that could change our lives". Fourth Estate, London 1994. ISBN 1-85702-302-1 (also published under the title "Alternative Science")

Dieter Britz's Review: The author contends that big science has distorted what science is all about, and now has its own useless momentum, and actively suppresses any new science that might act as a threat to ongoing big projects or challenge received wisdom. One of the recurring example cases is that of cold fusion, which Milton accepts as having been proven by 92 labs around the world. He believes that the hot fusion establishment is behind its suppression. Other cases are mentioned, such as the reaction to Velikovsky (whom he also seems to believe), Halton Arp (a good example, it seems to this abstracter) etc. He comes up with the interesting concept of experts in describing bad situations such as disease, but who cannot offer any solutions. Generally, big science is heavily (and convincingly) criticized. An excellent and thought-provoking book.

William Beaty's Review: "FORBIDDEN SCIENCE," excellent anti-skept book! finally discovered a copy of It sheds light on the "taboos" phenomena which rule modern science, and on the hidden battle going on between the skeptical orthodoxy and the frontier researchers. It contains details of numerous historical examples where the skeptics WERE wrong, where skeptical suppression really did greatly impede progress. It's quite well done and intelligently well argued (though I'm biased heavily in favor of unconventional research, so make your own decision!) I was glad to see details about the twisted thinking which goes into automatic rejection and ridicule of "obviously crazy" claims. In the section on the Wright Brothers, we hear that the Wrights conducted their later flight experiments in 1904 and 1905 in Dayton Ohio in an open field bordered by two main highways and a rail line. (cont. page 8)

Cavitation Induced MicroFusion

Russ George

Cont. page 1

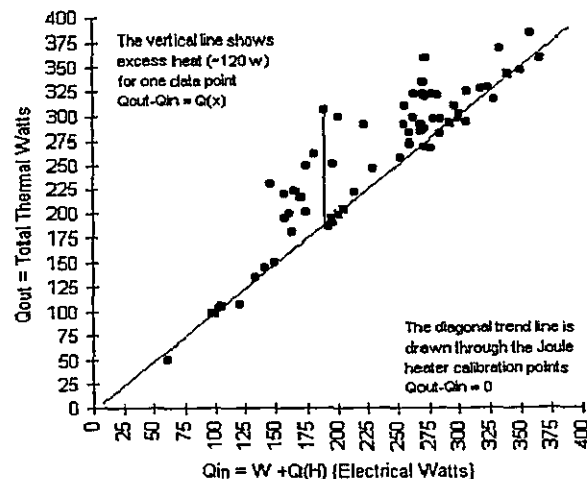


The evidence from this project demonstrates that substantial excess heat is highly reproducible but not with perfect predictability. The evidence for ⁴He shows it is present as a signature of the nuclear reaction(s) generated but is unlikely the product of a single principal heat producing reaction. Indeed the helium producing reactions must be relatively minor contributors to the anomalous heat in this series of experiments. Isotopic shifts in the Pd isotope ratios suggest a link to a reaction involving the entry of a hydrogen isotope into the large Pd nucleus. The precise nuclear reactions occurring in the experiments remain difficult to define and are the subject of an intensive ongoing search.

The first experiments conducted resulted in dramatic melting of palladium target foils (5 cm x 5 cm x 0.1 mm) while immersed in flowing D₂O and impinged upon by an acoustic field of approximately 2-3 watts per cm². Calorimetric studies revealed production of a large amount of excess power relative to the electrical input power consumed to produce the ultrasonic energy. Later experiments included analyses of gases from the helium tight stainless steel apparatus and revealed relatively large amounts of ⁴He as measured by 3 independent laboratories. While the ⁴He observed was a significant signal it was not in sufficient quantity to be commensurate with the excess heat observed using standard nuclear fusion reaction paths such as D+D → ⁴He (24 MeV). Experiments in this project were designed to repeat, with both E-Quest and SRI staff, previous experiments, as conducted by E-Quest at Los Alamos National Laboratory in 1993 and in E-Quest labs from 1989-present, and to search for coincident ⁴He. The experiments show reproducible large excess power (tens to hundreds of thermal watts) is generated in the device based on calorimetric measurement of total energy input and heat output. The figure shows a summary plot of data from the experiments conducted at SRI during the course of this project. The diagonal line is drawn through points determined via the use of a Joule heater and show measurement of heat in equal to heat out. Many of the experimental points which include some experiments with inactive ingredients fall on or near the Joule heater calibration confirming the case for no excess heat. However many points fall far to the left of the calibration line and demonstrate excess energy. The one point which is attached to a vertical line is used to illustrate the substantial excess power which in this case is about 160% of the input of just under 200 watts.

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ENGINEERING AND MATERIALS VIEW TOWARD COLD FUSION

Anomalous Isotopic Distribution

T. Mizuno Continued from page 5

are shown in Figure 3; Cr, Fe and Cu showed large shifts in isotopic ratios. Especially notable was the fact that no ^{64}Cu peak was observed. Except for a few cases, in general the isotope abundances are higher for odd mass numbers and lower for even ones, as compared with the natural ratios. In Figure 4, for mass numbers between 100 and 140, Cd and Xe were shown to have large shifts in abundances. In Figure 5, for mass number between 172 and 208, large shifts were seen for Hf, Re, Ir, Os and Hg. Generally speaking, heavier isotopes increased compared to the natural ratio, and lighter isotopes decreased. For example, Mg^{24} went from 79% down to 74%, while Mg^{25} went from 11% to 12%. This rule did not hold for some elements with few isotopes: Cu^{63} increased while Cu^{65} was absent; Ti^{47} increased while Ti^{48} decreased. The pattern is confusing for Cd^{111} to Cd^{114} : a slight decrease, a giant increase, a large increase and compensating drop in heaviest isotope, number 114. Data from some elements are not shown in Figures 3 to 5 because their peaks overlapped with those of other elements, or because they showed only a small difference with natural abundance. Neutron intensity and energy measurements were carried out simultaneously, in parallel. The neutron evolution rate was sporadic and weak, as previously reported (2), with levels of ~ 0.4 counts per second. No gases such as He, O_2 and Ar were detected. Excess heat generation was less reproducible, varying from 10^2 to 10^7 joules. (3)

Discussion Essentially the same phenomenon was confirmed five times with high reproducibility at high cathodic current density, above 0.2 A cm^2 . Current density ranged from 0.2 to 0.6 A/cm^2 . Different isotope distributions were obtained, depending on the current density. This will be described more fully in an upcoming paper. It can hardly be imagined that all of the elements found were impurities in electrolyte, electrode or cell. Even if we suppose that all impurities in the system accumulated in the cathode, the amount would be 10 to 100 times smaller than the total amount we detected. Furthermore, it is simply impossible to explain the shifts in the isotopic distribution. Hence, it must be concluded that some novel reactions occurred, resulting in the reactants which were found abundant in the electrolyte and electrode material. We assume the cathode palladium was the starting material for these reactions, but it is possible that impurities and other cell components such as Li, D_2O , Pd, Pt, K, Na, Ca, B, C, Ag and Fe may have provided the starting material for the nuclear reactions.

The palladium surface became rough and porous after several weeks of electrolysis, probably due to hydrogen attack. The current may have increased in such roughened surface areas, which would in turn cause a larger reaction and a higher concentration of the reaction products. Enyo (4-6) reported that the effective hydrogen pressure at the hydrogen evolving electrode depends upon the hydrogen atom recombination process which follows the water discharge process. The division of the overpotential between these two steps may be important. It was suggested that at Pd electrodes in alkaline solution the effective hydrogen pressure may rise as high as 10^6 atm (7,8). There may be further increases at local points on a heterogeneous surface. One may even speculate that the hydrogen isotope nucleus sometimes closely approaches the medium nucleus (9). An estimate by Nernst Equation indicates that $5 \times 10^4 \text{ atm}$ of pressure may be realized at 140 mV of overpotential at the electrolysis current density of 200 mA cm^2 on a flat palladium surface. (10) Furthermore, the pressure distribution depends on the roughness of the surface, because local current density and the partition of overpotential components may vary with roughness. There are several reports (11-13) of evolution of elements by electrolysis. However, only a few (14,15) demonstrated shifts in isotopic abundance. For a gold electrode (16), which also generated various elements by cathodic electrolysis different isotopic distributions were seen. In this case also, the reaction sites were unevenly distributed on the surface. Typically, the active points may have occupied about 10^{-6} cm^2 areas and numbered 10^4 to 10^5 per cm^2 at the surface. Thus, the current is likely to be concentrated at localized points 10 to 100 times higher

Cavitation Induced MicroFusion

Cont. page 6

Melting and damage to the palladium targets was a frequent occurrence providing additional evidence of anomalous large heating events within the palladium lattice. Excess power generation was more or less continuous over the operation of experiments lasting typically 24 hours. Nearly 1/3 of the experiments failed to produce excess heat and represent ideal "controls" which are in close agreement with the Joule heater calibration. Experiments showing excess power ranged from a few tens of watts to over 100 watts measured to an accuracy of $\sim 5\text{-}10\%$. The measured ^4He is found to reveal a trend of increasing ^4He associated with increasing excess heat, power input, and experiment duration. The ^4He was found in smaller amounts (2×10^{14} - 10^{15} atoms vs. 10^{17} - 10^{18} atoms) than in previous experiments but is shown in context of a demonstrated low background. In earlier experiments the helium produced was shown to be isotopically unique in that it was not associated in the expected or natural ratio with ^3He or ^2Ne . Isotopic analysis of the helium produced in the experiments described herein was not within the scope of this project. Studies of the palladium target metal using high resolution dynamic and time of flight SIMS techniques have revealed anomalies of isotope ratios in the palladium isotopes and a variety of other isotopes in the targets. This evidence is strongly suggestive of particular nuclear reactions but confirmation awaits additional studies using SIMS and other isotope ratio analytical techniques. ➤

than average. Such high pressure can induce neutrons to enter into heavy nuclei and successively form an unstable heavy nucleus (17). If the reactions start from palladium as the electrode material, fusion and fission may take place simultaneously. Atomic numbers of 20, 28, 50 and 82 are called magic numbers. Here, ^{136}Xe nuclei are abnormally increased, and hence the number of neutrons changes to magic number 82, and especially stable nuclei are selectively produced. In this way, the isotopic distributions of the products after electrolysis can be explained in terms of the difference of binding energy of the nuclei. It must be admitted that these reactions have no solid, detailed theoretical basis yet, but in broad terms this can explain most of the elements which were observed. One may also imagine that as such transmutation reactions were presumably taking place during the electrochemical process, they are likely to be connected with other phenomena such as hydrogen embrittlement and local corrosion.

Conclusion Anomalous evolution of various elements in palladium electrodes was confirmed after high current density cathodic electrolysis under high temperature and pressure for a long time. The following conclusions were drawn: (1) The mass number of the evolved elements were distributed roughly in three groups: 20 to 28, 46 to 54 and 72 to 82, with the amounts, respectively, $>50\%$, 10% and $<5\%$ compared to palladium.

(2) These evolved elements were found on the surface, and also in the bulk of the electrode in amounts 10 to 100 times smaller.

(3) Many evolved elements accumulated in holes and cracks on the electrode which formed during electrolysis.

Continued on page 8

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Anomalous Isotopic Distribution

T. Mizuno Continued from page 7

(4) Some of the evolved elements have isotopic distributions drastically different from natural ones, especially for Cu, Br, Xe, Pd, Cd, Hf, Re, Pt, Ir and Hg.

(5) Gaseous Xe was noteworthy because it was so abundant. Xenon is particularly unlikely to be a contaminant because metals do not absorb noble gases, and because the cathode was degassed in a vacuum at 473 deg K for 20 hours.

(6) The elements in the bulk layer changed in concentration with depth and showed shifts of isotopic distribution.

(7) Light elements such as O, C, Ca, Na, Mg and Al showed small isotopic shifts.

(8) Ni and Co were also confirmed but their isotopic distribution could not be measured because their SIMS peaks overlapped with those of other elements.

(9) The isotopic distributions of Pd and Pt were also shifted.

(10) We conclude that nuclear reactions must have occurred during the electrochemical process.

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William Beatty's REVIEW of

"FORBIDDEN SCIENCE" (continued from page 6)

They gave rides to railway employees and managers, the field was donated by a Dayton bank president, and the local Dayton newspapers received so many inquiries from witnesses that the papers complained about the nuisance. Yet neither the *Dayton Journal* nor the *Daily News* carried a single newspaper story, and neither looked into the reports! In an interview about this, one of the editors stated "We just didn't believe it." And then in 1906, the *Scientific American* published an article ridiculing the alleged flights the Wrights claimed to be making as hoaxes, arguing that, had the flights been real, they would have been widely publicized in the newspapers! The book also goes into interesting details on believer/skeptic controversies both historical and contemporary, effects of beliefs on behavior among scientists, questionable CSICOP practices, scientific taboos and the forces supporting them, and lots of interesting 'dirt' that the staunch Hyper-skeptic would probably prefer remains forgotten. Oh, it also features a good bibliography and chapter references for those who'd like to check out the papers and publications.

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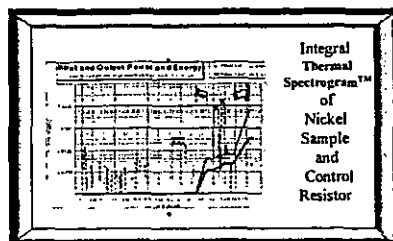
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Rothwell's ILENR2 Summary

Cont. from page 2

Miley is not the first to report metal transmutations in cold fusion cathodes. I believe that honor goes to Mizuno, who is usually years ahead of everyone else. But Miley's data is the most extensive, and his experimental technique and exposition are brilliant.

Mizuno [2] - The cathode is carefully prepared, and cleaned by being heated to 200 deg C for 16 hours. The levels of contamination measured in the cathode and cell components are carefully measured before the experiment, and this data proves that contamination could not possibly account for the elements found in the cathode. One of the cathodes used in this work underwent heat after death, in the first recorded instance of this in the literature. The Delta T temperature spontaneously rose 30 deg C and remained hot for three days.

Dash [3] - Dash has been working with Ti cathodes and slender resources. He reports excess heat and transmutations. He uses two air-cooled static calorimeters in series, attached to a data acquisition computer. One cell contains a blank, one a Ti cathode. The electrolyte is 0.99 mole D₂O and 0.01 mole H₂SO₄. These are closed cells with recombiners. The wires leading into the cells for electrolysis and the sensors pass through wax, which is not a very good method of sealing a closed cell. The cells weigh a little less after experimental runs because some gas is lost through the wax. In a typical excess heat run, the cold fusion cell consumes 20% less energy than the control, yet it is 4 deg C hotter. This represents a 1.2 watt excess with a 0.6 gram cathode. After he washes away galvanized contamination from the surface with an ultrasonic jewelry cleaner in de-ionized water, Dash finds that most of the surface highly pure Ti. This shows the cleaning method works well. But in the material that had erupted from below he found a mixture of different elements, which were surprising unevenly distributed within the ejecta. The SEM photos Dash showed during his lecture did not clearly show that the lumps on the surface of the cathode were ejecta; they looked like clumps of debris that had been deposited onto the surface. However, evidence in his Fusion Technology paper demonstrates that it comes from inside. He said that Ti damages more easily than Pd, and it is highly reactive.

Swartz [6] - Swartz described meticulous multi-ring calorimetry of a unique design, featuring multiple envelopes within envelopes. This allows high accuracy while persevering the isolation and cleanliness necessary to finding transmutations, where Miley and Claytor trade off calorimetric precision for cleanliness. Swartz's calorimetry is as conservative. He measures 1°V input and makes no adjustments for gas losses. His electrochemistry is unique in many ways: he uses no salts in the electrolyte, and he reports some of his best excess heat results with natural light water and a gold cathode, which produces 3 to 6 times input. Swartz hypothesizes that some nuclear ash might be in the form of hydrogen transmuted into deuterium, in a multi-body reaction. This would be extremely difficult to find because deuterium is not radioactive and because there is a great deal of it the ordinary water electrolyte to start with. A great deal, that is, compared to the amounts that would be synthesized in a nuclear reaction. It would take a megajoule to increase the deuterium content of the water by a few percent. He plans to verify the hypothesis with new in-situ spectroscopes he is now installing. I wonder if it would help if he used purified light water. That technique is difficult, because pure light water rapidly absorbs deuterium from the atmosphere when it is exposed to air. Swartz describes what he calls the "π-notch" graph of excess heat with Ni H₂O. You get the most heat with low current density. As current increases, the heat peaks, and it declines as a percent of total input. With Pd, higher current density usually increases loading, which increases the heat geometrically. After Swartz spoke, Miley suggested that perhaps at higher power endothermic reactions predominate, producing more transmutations but less heat. Swartz thought this is an interesting idea.

Claytor [12] - Claytor continues to report progress in tritium generation by plasma discharge (sparking) through deuterium gas. His best results to date have been with palladium alloys Pd (90%) Rh (5%) Co (5%). In 12 runs this worked well four times. When he adds a small amount of CO₂ (<1%) to the deuterium gas, the plasma wire often soaks up a larger volume of gas more quickly, and it produces more tritium. Most impurities poison the reaction, but some enhance it tremendously.

SEM photos of many cathode surfaces showed eruptions of material from inside, looking like the surface of the moon. [2, 3] It was this damaged material and the areas around it that was transmuted. In cases where the transmutations occurred well below the surface the material was not ejected but damage was seen when the outer layers were removed. [5] The transmuted elements were not uniformly distributed. Areas with no damage often remained highly purified metal (Pd, Ni, Ti or Au). This is a large effect, easily detected. In many cases, large amounts of the metal were transmuted. Miley [1] reported up to 40% of the total metal was transmuted, where the starting metal was better than 99% pure. Minevski said that in the layers where transmutations occurred, up to 70% of the material was no longer palladium. Mizuno reported three peaks of new elements: light, medium and heavy elements. Miley said he saw the same pattern, with the three peaks moving up or down the spectrum depending on the metal he started out with (Pd or Ni).

1. G. Miley (University III), Nuclear Reaction in Palladium-Hydrogen System
2. T. Mizuno (Hokkaido Nat. University), Reaction Products Induced by Isotopic Changes of Electrolysis
3. J. Dash (Portland State University), Excess Heat and Unexpected Elements from Electrolysis of Acidified Heavy Water with Titanium Cathodes
4. S. Szpak (COSC, Navy), Nuclear and Thermal Events Associated with Pd+D Codeposition

5. Z. Minevski (Lynntech), Two Zones of "Impurities" Observed After Prolonged Electrolysis of Deuterium on Palladium
6. M. Swartz (JET Energy Tech.), Deuterium Production and Light Water Excess Enthalpy Experiments Using Nickel Cathodes
7. R. George (E-Quest Sciences), Isotopic Ratio Anomalies Induced in Palladium by the Application of Intense Ultrasound
8. H. Fox (Fusion Info Center), Low-Energy Nuclear Reactions in an Electrolytic Cell

9. G. H. Lin (Texas A&M University), Anomalous Radioactivity and Unexpected Elements as a Result of Heating Oxide Mixtures
10. T. Grotz (Wireless Engineering), Synthesis of Iron via Arc Discharge through Activated Carbon
11. K. Shoulders, Observations on the Role of Charge Clusters in Nuclear Cluster Reactions
12. T. Claytor (Los Alamos National Laboratory), Tritium Production From Palladium and Palladium Alloys

History Repeats Itself? Diocletian and Texas A&M

Alchemy, ancient art practiced especially in the Middle Ages, devoted chiefly to discovering a substance that would transmute the more common metals into gold or silver and to finding a means of indefinitely prolonging human life. The birthplace of alchemy was ancient Egypt, where, in Alexandria, it began to flourish in the Hellenistic period; simultaneously, a school of alchemy was developing in China. The Roman emperor Caligula is said to have instituted experiments for producing gold from orpiment, a sulfide of arsenic, and the emperor Diocletian is said to have ordered all Egyptian works concerning the chemistry of gold and silver to be burned in order to stop such experiments. The earliest authentic works extant on European alchemy are those of the English monk Roger Bacon and the German philosopher Albertus Magnus; both believed in the

possibility of transmuting inferior metals into gold. The Italian Scholastic philosopher St. Thomas Aquinas, the Catalan churchman Raymond Lully, and the Benedictine monk Basil Valentine (flourished 15th century) also did much to further the progress of chemistry, although along alchemical lines, in discovering the uses of antimony, the manufacture of amalgams, and the isolation of spirits of wine, or ethyl alcohol. (Encarta Encyclopedia)

WORLD WIDE WEB SOURCES ON COLD FUSION

- **COLD FUSION TIMES** home page:
<http://world.sid.com/~mica/cft.html>
- John Logajan's home page:
<http://www.skypoint.com/members/jlogajan>
- *Infinite Energy* home page:
<http://www.mit.edu:8001/people/rci/CFdir/CFhome.html>
<http://ourworld.compuserve.com/homepages/JedRothwell/>

ILENR2 Meeting (September 1996, Texas) - Attendants listen to reports



OP-ED COLUMN --

A memo written by Distinguished Professor John O'M. Bockris and distributed at The Second International Low Energy Nuclear Reactions Conference (ILENR2) September 13, 1996 College Station, Texas.

WHY WE ARE NOT MEETING AT TEXAS A&M UNIVERSITY

I applied to the Department Head, who had sanctioned last year's Meeting. He referred it to the newly formed Chemistry Department Advisory Committee. I supplied each member of the Committee with a copy of Storm's 1996 review before the Meeting at which a vote was to be taken. I briefly addressed the Committee at the Meeting.

The vote was 12 to zero negative, no abstentions.

I asked the Department Head in two memos why he had voted against having the Meeting. He did not reply to this question. I spoke with a Member of the Committee and he said that the opinion was that the field did not exist, that any suggestion that it did must constitute fraud and the idea of having a Meeting to discuss it, a joke. I sent him references to the Reifenshweiller experiments described in Physics B, and referred to publications exceeding 1000 in number. He did not reply.

Texas A&M ranks number 2 in the country in respect to the facilities for nuclear research within a University, (e.g., as in the Cyclotron Institute).

*J. O'M Bockris
6 September 1996*

TO COLDLY GO WHERE NO ONE HAS GONE BEFORE

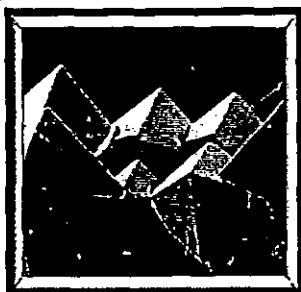
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MORE REPORTS:

News and Developments from California, Washington, D.C., Illinois, Massachusetts, New Mexico, Rhode Island, Texas, Utah, China, England, France, India, Japan, Netherlands, Russia, Sri Lanka, United Kingdom. Plus copious material science and engineering news.

This issue of the **COLD FUSION TIMES** features reports on nucleosynthesis. Inside are ILENR2 meeting summaries, and other reports showing monotonic increasing depth alterations in isotopic distributions. Here Dr. Lin and Bockris, Department of Chemistry, Texas A&M University, report anomalous radioactivity and the *de novo* appearance of unexpected elements in some of their experiments.

Anomalous Radioactivity and Unexpected Elements Upon Heating Inorganic Mixtures

G. H. Lin and J. O'M. Bockris

College Station, TX - Rutherford [1-2] at Cambridge in 1919 achieved the first "artificially induced transmutation" by

bombarding nitrogen with alpha particle from radium to produce oxygen. This achievement founded high energy physics but it also create a mind set, breaking a nucleus needed a million times more energy than is given out in a chemical reaction. The first reported cold transmutation in modern times is that of Ramsay et. al., who claimed the transmutation of other elements into helium and neon, reported by J. J. Thomson [3]. Thomson did a similar experiment as that of Ramsay, but he was uncertain whether the liberation of rare gases upon bombardment in a discharge tube is due to the fact that the gases are earlier absorbed from air or whether they are formed by means of transmutation. Kervran in 1955 did some systematic studies in biological systems where he described the conversion of Na to K [4]. George Oshawa et al [5] and Michio Kushi [6] in 1964 made series of experiments on the transmutation of elements. These experiments were performed at normal temperature and pressure through electrochemical methods. The experiments described by Oshawa et al for the production of Fe from C and O allow one to reproduce the experiments and has been repeated recently [7-9]. Many recent reports [10] are now appearing in the literature which suggest low energy nuclear conversion of elements.

The recovery of precious metal (noble metals) using fire assay-ing has been used for a long time and a number of variations of this technique have been proposed.

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